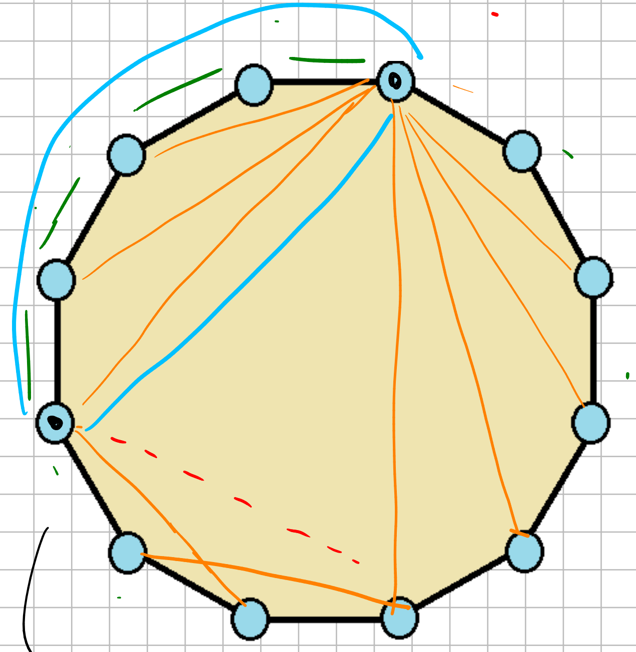
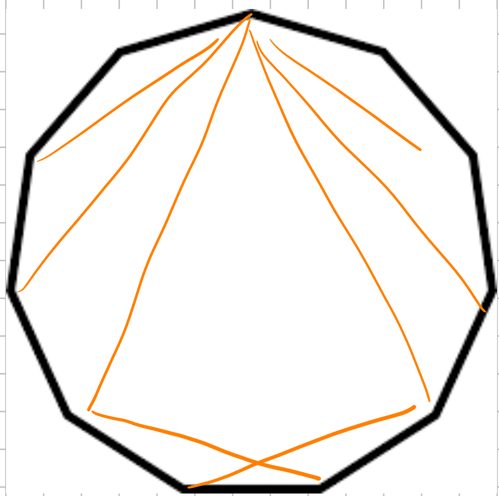
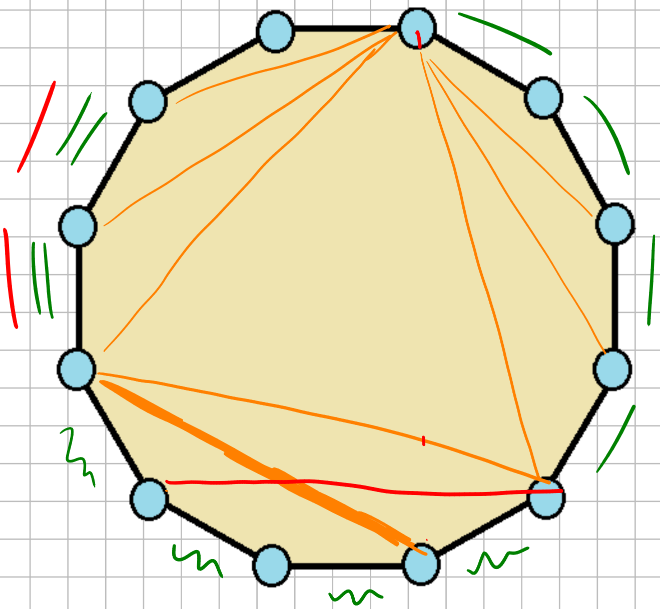
$k=8$

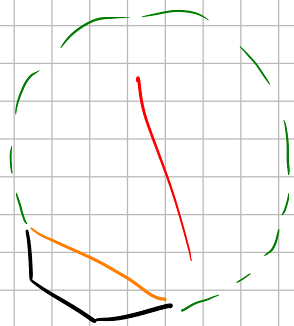




survives $k=1,2,3$

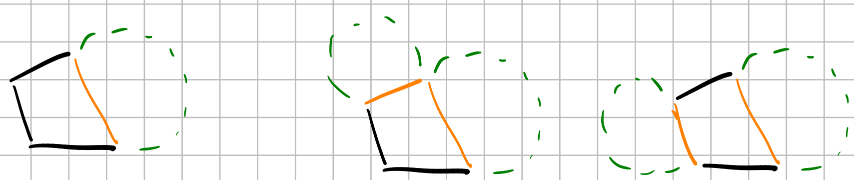
but not 4

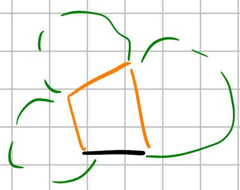
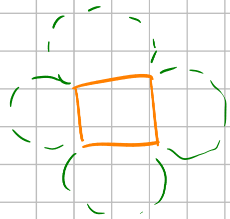




Prop. a diagram that is non-local and 1-step removed from a triangulation (we can remove 1 chord and then nothing changes) is always killed by some k-zero.

Proof once chord is removed, there are 5 possible things that can happen:

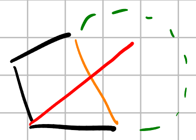
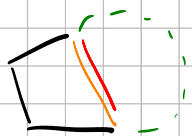


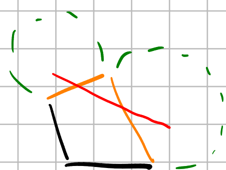
once we put back the crossing chord, some n-zero always kills the diagram

- the crossing cannot be only inside the black/orange 4-gon (otherwise is not a crossing)
- if only inside one of the green regions we fail the 1-zero test

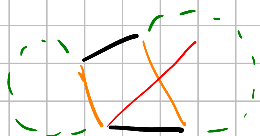
→ must cross orange

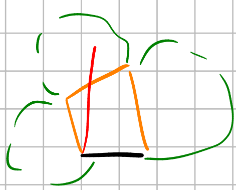
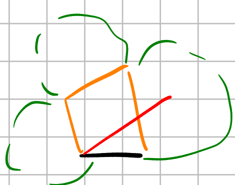
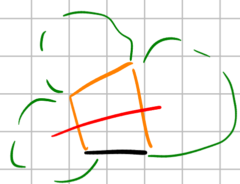
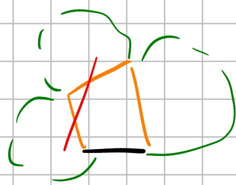
both fail 1-zero

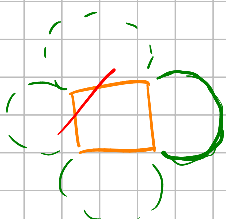
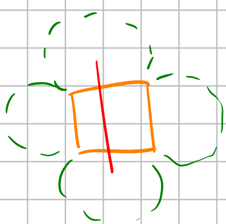
 

both fail 1-zero

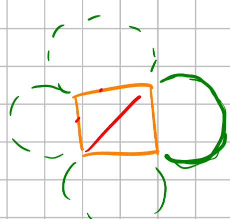
 

both fail 1-zero

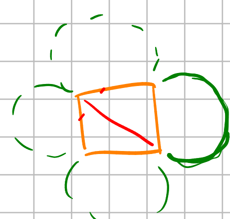
   

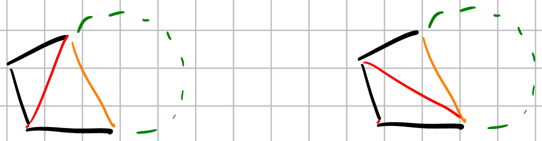
 

↳ fails because it needs



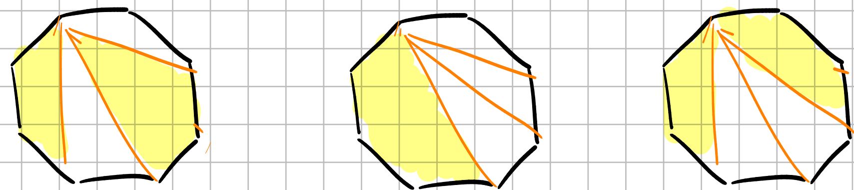
or





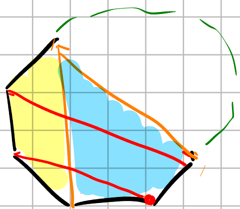
2 steps removed

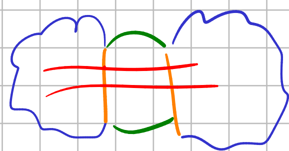
ex:



both chords need to cross something that is already there

→ each shaded region needs to be crossed (otherwise some κ-zero fails)



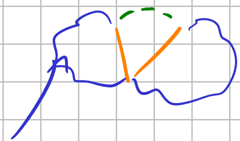


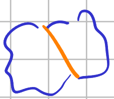
blue blobs are cases from step 1

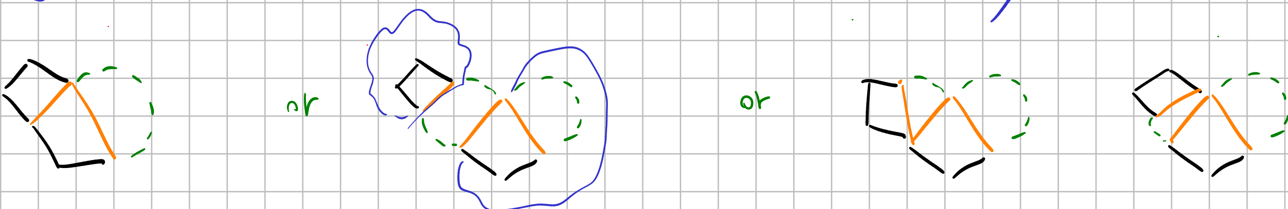
each blob has a 4-gon inside (with orig chord) and from earlier step we know that the term dies if we don't add an inner chord to each 4-gon.

→ so we need to make a Δtion

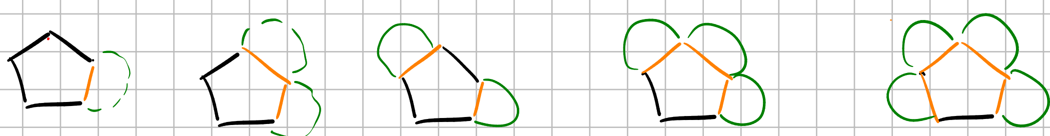
ex:

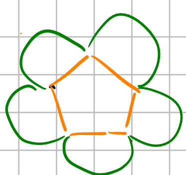






what about pentagon?

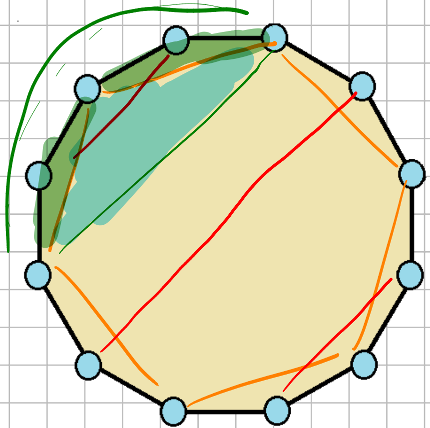
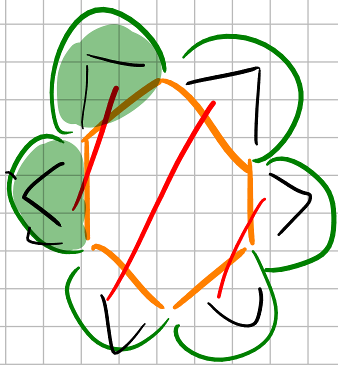
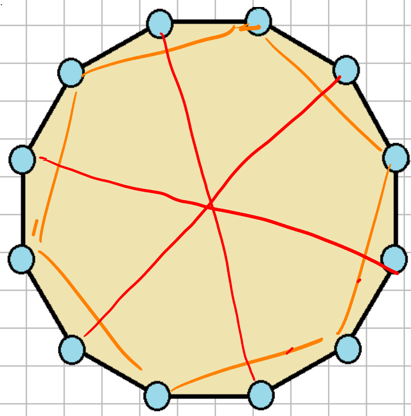




add 2 back in

if one of them is an inner chord

then we know from 4-gon case that the other is also an inner chord

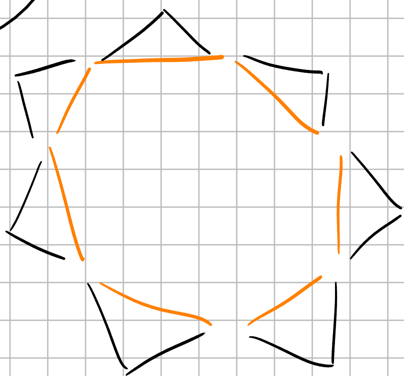
n	# blobs that we can cross
5	4
6	6
7	8
8	10

test: look at all digons of the form

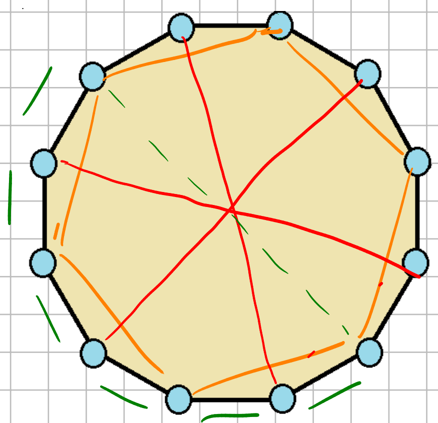
$2n - 6$ To put

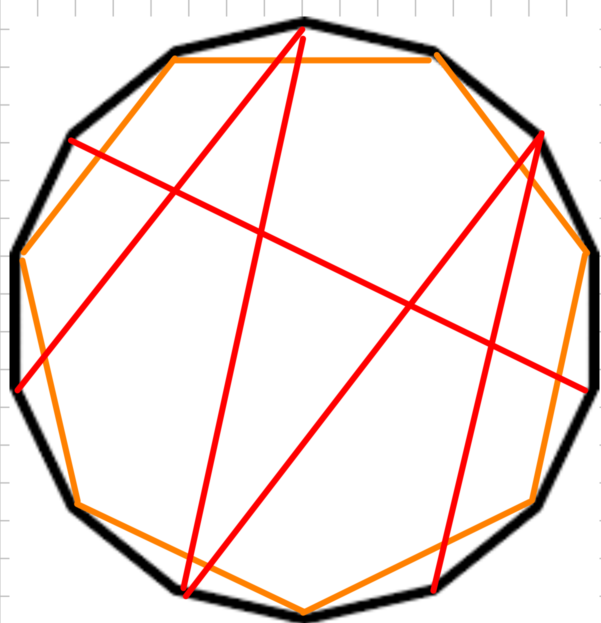
n

$n - 6$ left



to test numerically





Counter-example to current approach

so we need the counts we ignored:

k-zero

$$X_{2_0} = X_{k+1_0} + X_{2k}$$

$$X_{3_0} = X_{k+1_0} + X_{3k}$$

$$\vdots$$

$\boxed{\text{2-zero}}$

$$X_{35} = X_{15} - X_{14}$$
$$\vdots$$
$$X_{3\,n-1} = X_{1\,n-1} - X_{14}$$
$$X_{3n} = -X_{14}$$

$$X_{2j} = X_{3j} + X_{24} \qquad j = 5, \dots, n$$

free: X_{13}

X_{46} X_{47} -- X_{4n}

X_{57} -- X_{5n}

$\vdots$

$X_{n-2\,n}$

things we cannot not send to ∞: X_{14}, X_{3n}

from each eq. we want to send everything but 1 thing to ∞

$$X_{25} = X_{35} + X_{24}$$
$$X_{26} = X_{36} + X_{24}$$
$$\vdots$$
$$X_{2n} = X_{3n} + X_{24}$$

if X_{24} is free : X_{3j} not always go to ∞

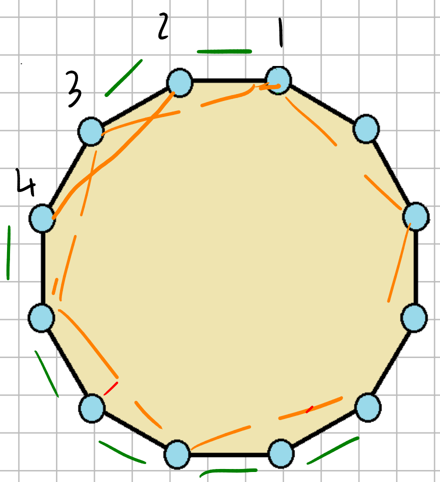
$\rightarrow$ $\boxed{24, 15, 16, \dots, 1n-1 \text{ as independent}}$

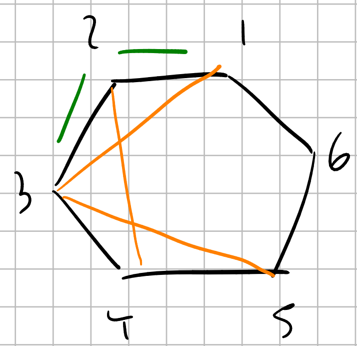
$$X_{36} = X_{16} - X_{14}$$
$$\parallel$$
$$X_{26} - X_{24}$$

if $X_{24} = X_{14} \to \infty$

- either $X_{35} \to \infty$ or $X_{25} \to \infty$
- either $X_{36} \to \infty$ or $X_{26} \to \infty$

X_{2n} can be finite





$n=6$

$$X_{35} = X_{15} - X_{14} \qquad X_{25} = X_{35} + X_{24}$$
$$X_{36} = -X_{14} \qquad X_{26} = X_{36} + X_{24}$$

sending $X_{14} \to \infty$: $X_{36} \to -\infty$ $\quad (X_{26} = X_{24} - X_{14})$

(A) choose X_{15} finite so $X_{35} \to -\infty$

- $X_{24} = a - X_{33}$ so X_{25} is finite and $X_{24} \to +\infty$

and then $X_{26} = -X_{14} + X_{24} = -X_{15} + X_{25}$

→ this is not good! because our framework needs the free things to be independent

if X_{15} finite we must choose X_{24} finite and $X_{25} \to \infty$

what about X_{26}? also $X_{26} \to \infty$

(B) if X_{35} finite: $X_{15} \to +\infty$

must have $X_{24}, X_{25} \to \infty$ otherwise 3 related independents

if $X_{24} \to \infty$ and $X_{36} \to -\infty$ we could have

X_{26} finite